Multiple-choice questions

|!|BCEMC0080000001|!|

Which of the following is a structural feature of air sacs for gas exchange?

A. They are spherical in shape so as to increase surface area.
B. Their oxygen concentration is higher than air in the atmosphere to increase the rate of diffusion.
C. Their walls are only one-cell-thick for rapid diffusion of gases.
D. They are small in size so the gases inside them are more concentrated.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000002|!|

Where does gas exchange take place in humans?

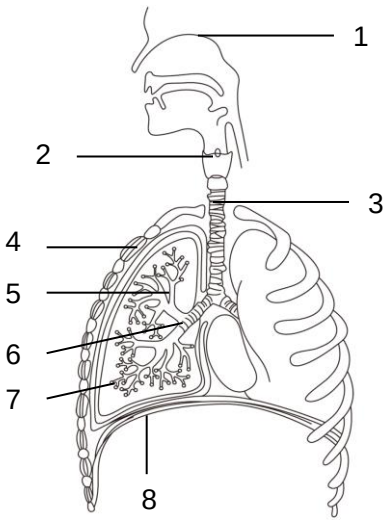
A. nasal cavity
B. trachea and bronchi
C. walls of bronchioles
D. walls of air sacs

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000003|!| ★

The diagram below shows different parts of the human breathing system.



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Cilia can be found in parts

A. 1, 2 and 3.
B. 1, 3 and 6.
C. 2, 3 and 5.
D. 2, 3 and 6.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000004|!|

Which of the following muscle actions causes air to be expelled from the lungs?

	Intercostal muscles	*Diaphragm muscles*
A.	contract	contract
B.	contract	relax
C.	relax	contract
D.	relax	relax

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000005|!| ★

Which of the following statements correctly shows the relationship between respiration and ventilation in humans?

A. Ventilation and respiration are the same process.
B. Respiration releases carbon dioxide which is removed from the lungs by ventilation.
C. Respiration uses oxygen which is carried to body cells by ventilation.
D. The higher the respiration rate, the lower is the ventilation rate.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000006|!|

Which of the following is the correct path of air passing out of the lungs during exhalation?

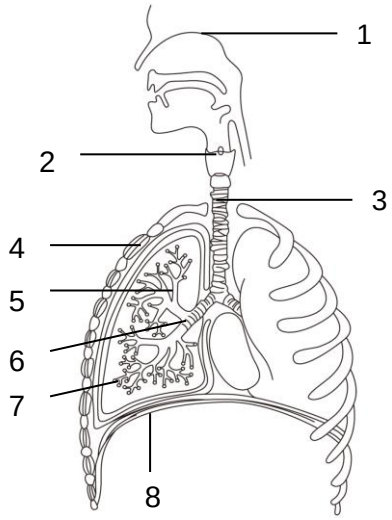
A. air sacs → bronchi → bronchioles → trachea
B. air sacs → bronchioles → bronchi → trachea
C. air sacs → trachea → bronchi → bronchioles
D. trachea → bronchi → bronchioles → air sacs

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000007|!| ★

The diagram below shows different parts of the human breathing system.



The gills in a fish perform the same function as the part labelled

A. 4.
B. 5.
C. 7.
D. 8.

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0080000008|!| ★

The carbon dioxide content of the exhaled air in humans is about

A. 1%.

B. 4%.

C. 20%.

D. 78%.

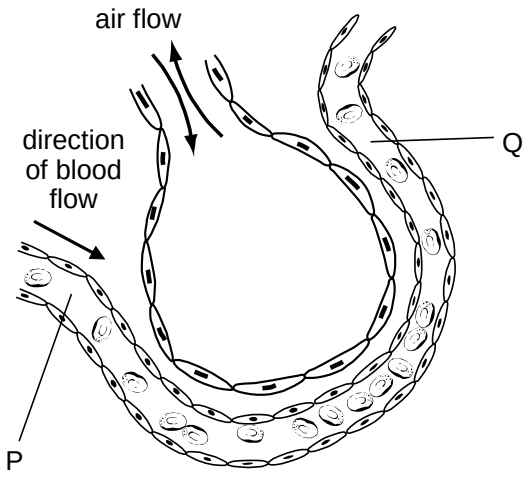
□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000009|!| ★

The diagram below shows an air sac surrounded by a blood capillary.



Gas exchange occurs when blood flown from P to Q. Which of the following combinations correctly compares the concentrations of gases at P and Q?

	Oxygen concentration at P	***Carbon dioxide concentration at Q***
A.	higher	higher
B.	higher	lower
C.	lower	higher
D.	lower	lower

□

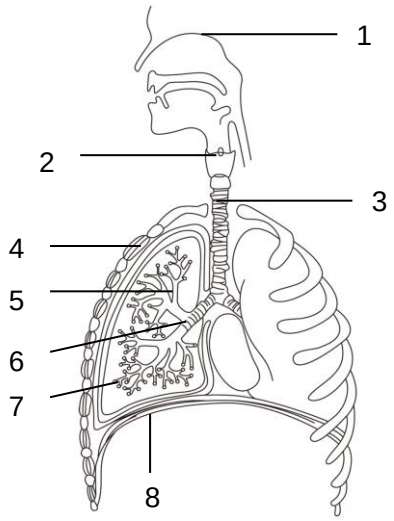
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000010|!|

The diagram below shows different parts of the human breathing system.



Which structures are responsible for the increase in thoracic volume during ventilation?

A. 4 and 7
B. 4 and 8
C. 5 and 7
D. 7 and 8

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000011|!| ★ **Cross topic**

Which of the following statements about the oesophagus and trachea is ***incorrect***?

A. The passing of food down the oesophagus is active while the passing of air down the trachea is passive.
B. The oesophagus is made up of layers of muscles while the trachea is mainly made up of C-shaped rings of cartilage.
C. Both the oesophagus and trachea open to the larynx.
D. During swallowing, the epiglottis automatically covers the entrance to the trachea. This stops food from entering the trachea.

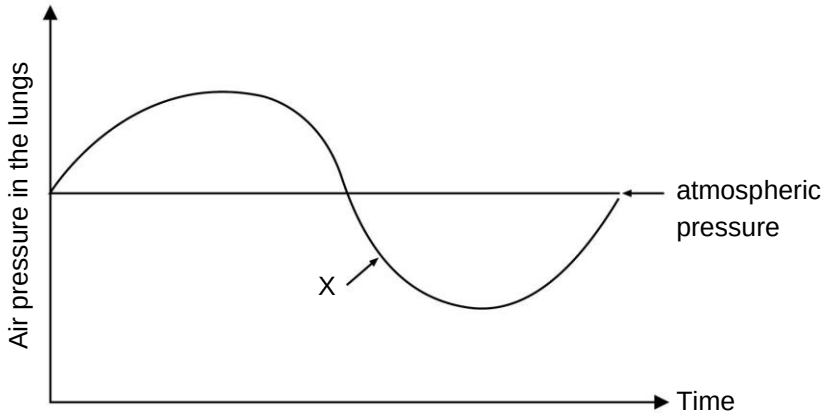
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0080000013|!| ★

The graph below shows the change in air pressure in the lungs of a man in one normal breath.



What of the following correctly shows the states of the diaphragm muscles and rib cage at time point X?

	Diaphragm muscles	***Rib cage***
A.	contract	moves upwards
B.	contract	moves downwards
C.	relax	moves upwards
D.	relax	moves downwards

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000014|!| ★

Cigarette smoke contains tar that will deposit on the inner surface of air sacs. This will cause

A. an increase in the oxygen content in blood.
B. a decrease in the diffusion distance of gases.
C. a decrease in the amount of oxygen entering the lungs.
D a decrease in the efficiency of gas exchange.

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0080000015|!| ★

How is carbon dioxide produced by body cells transported in the blood of humans?

(1) as hydrogen peroxide in the blood plasma
(2) as hydrogencarbonate ions in the blood plasma
(3) combines with haemoglobin in the red blood cells

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000016|!| ★

Haemoglobin binds most readily with

A. oxygen.
B. carbon dioxide.
C. carbon monoxide.
D. nitrogen.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000017|!| ★

Which of the following statements about haemoglobin is ***incorrect***?

A. Haemoglobin contains iron.
B. Haemoglobin cannot combine with carbon dioxide.
C. The uptake and release of oxygen by haemoglobin is reversible.
D. Haemoglobin gives the blood a red colour.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0080000018|!| ★

In the blood, most carbon dioxide is carried

A. in the form of dissolved carbon dioxide in the red blood cells.
B. in the form of dissolved carbon dioxide in the blood plasma.
C. in the form of hydrogencarbonate ions in the red blood cells.
D. in the form of hydrogencarbonate ions in the blood plasma.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000019|!| ★ **Cross topic**

Which of the following statements about gas exchange in organisms is ***incorrect***?

A. Turtles carry out gas exchange with their gills.
B. In *Paramecia*, gas exchange takes place across their body surface by diffusion.
C. Whales carry out gas exchange with their lungs.
D. Most terrestrial plants carry out gas exchange through stomata.

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0080000020|!| ★

Which of the following statements about the breathing system is ***incorrect***?

A. The air sac receives blood from the pulmonary artery.
B. There are cartilages in the walls of bronchioles.
C. Cilia are absent on mucus-secreting cells.
D. The open end of the C-shaped ring of cartilage surrounding the trachea is adjacent to the oesophagus.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0080000021|!| ★

Which of the following statements about the pulmonary blood vessels is correct?

A. Pulmonary vein carries blood to the lungs.
B. Pulmonary vein carries deoxygenated blood.
C. Pulmonary artery carries blood to the heart.
D. The carbon dioxide content in the blood inside the pulmonary artery is higher than that the pulmonary vein.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000022|!| ★

Which of the following gases has the greatest percentage change between inhaled and exhaled air?

A. nitrogen
B. carbon dioxide
C. oxygen
D. helium

##B##

Skill assessed	Mark
Mathematic skills	1

|!|BCEMC0080000023|!|

Cartilages are ***not*** present in the

A. larynx.
B. trachea.
C. bronchi.
D. bronchioles.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000024|!|

Gas exchange can take place

(1) in the skin of frogs.
(2) in the gills of fish.
(3) in the air sacs of snakes.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000025|!|

Which of the following happens during exhalation?

A. The intercostal muscles contract.
B. The diaphragm becomes dome-shaped.
C. The air pressure inside the lungs is lower than the atmospheric pressure.
D. The volume of the thoracic cavity increases.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000026|!|

Which of the following does ***not*** occur during inhalation?

A. The air pressure inside the lungs is higher than the atmospheric pressure.
B. The volume of the thoracic cavity increases.
C. The rib cage moves upwards and outwards.
D. The diaphragm becomes flattened.

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000027|!|

What is the function of the fluid between the pleural membranes?

A. To moisten the incoming air.

B. To trap bacteria in the incoming air.

C. To reduce friction between the lungs and the ribs during breathing.

D. To nourish the lungs.

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000028|!|

Which of the following combinations correctly describes the composition of the exhaled air when compared with the inhaled air?

	Oxygen	*Carbon dioxide*	*Water vapour*
A.	more	less	the same
B.	less	more	the same
C.	less	more	more
D.	more	less	more

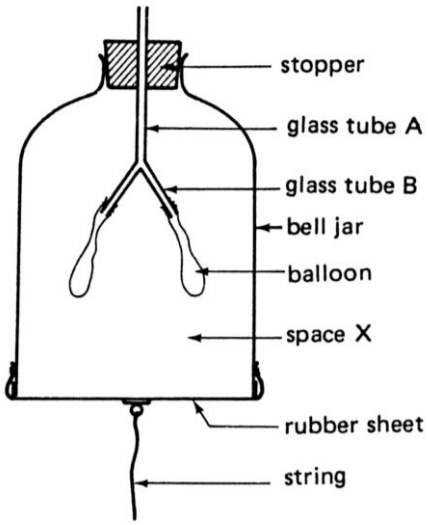
□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000029|!|

The figure below shows a model of the human breathing system.



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Which of the following combinations correctly shows the structures of the breathing system that correspond to the parts of the model?

	Glass tube A	*Glass tube B*	*Rubber sheet*
A.	bronchus	bronchiole	diaphragm
B.	trachea	bronchus	pleural membrane
C.	oesophagus	bronchus	diaphragm
D.	trachea	bronchus	diaphragm

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000030|!| ★

Mist is formed when a person breathes on a mirror. Which of the following is/are the correct explanation(s) of such phenomenon?

(1) Exhaled air is warmer than the mirror.
(2) Exhaled air is saturated with water vapour.
(3) Exhaled air has more carbon dioxide than the atmospheric air.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0080000031|!|

What are the effects brought about by the movements of the diaphragm and the ribs during exhalation?

	Volume of the thorax	*Pressure inside the thorax*
A.	increases	increases
B.	decreases	increases
C.	increases	decreases
D.	decreases	decreases

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0080000032|!| ★

Which of the following structures contain cartilage?

(1) air sac
(2) epiglottis
(3) larynx

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000033|!|

Which of the following are found in the inner wall of the trachea?

(1) ciliated epithelial cells
(2) mucus-secreting cells
(3) cartilage

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000034|!| ★

Which of the following statements about the lungs is/are correct?

(1) Lungs control the rate of breathing.
(2) Lungs are spongy due to the presence of numerous air sacs.
(3) Lungs can contract and relax during ventilation.

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A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000035|!|

Which of the following statements about the human breathing system is ***incorrect***?

A. The C-shaped cartilage rings in the wall of the trachea can prevent the trachea from collapsing.
B. The inner wall of the nasal cavity is lined with ciliated epithelial cells.
C. The intercostal muscles and the diaphragm muscles are involved in ventilation.
D. The bronchioles are supported by cartilage.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000036|!| ★ **Cross topic**

Which of the following are the functions of the rib cage?

(1) It protects the lungs.
(2) It helps in ventilation.
(3) It stores calcium and phosphate.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000037|!|

The large number of air sacs in the lungs is beneficial to gas exchange because

A. this can provide a large surface area for diffusion of gases.
B. this can hold a large volume of air for gas exchange.
C. this can provide a short distance for diffusion of gases.
D. this can maintain a steep concentration gradient of gases between the blood and air sacs.

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000041|!|

Which of the following combinations shows the approximate percentage of oxygen and carbon dioxide in the exhaled air?

	Oxygen (%)	*Carbon dioxide (%)*
A.	16	4
B.	16	0.03
C.	21	4
D.	21	0.03

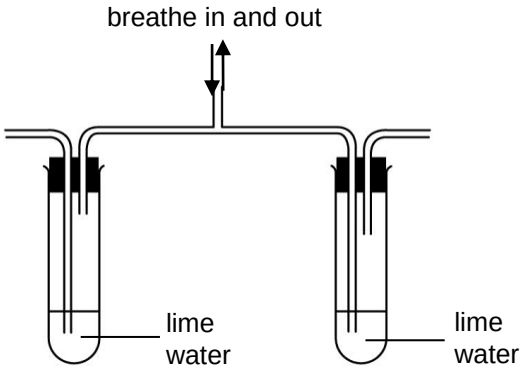
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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000042|!| ★ **SI**

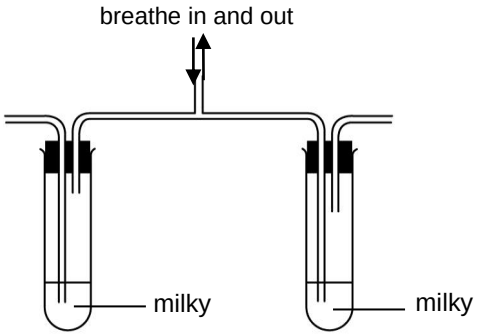
In an investigation to compare the carbon dioxide contents in inhaled and exhaled air, a student breathes in and out through an apparatus as shown below.



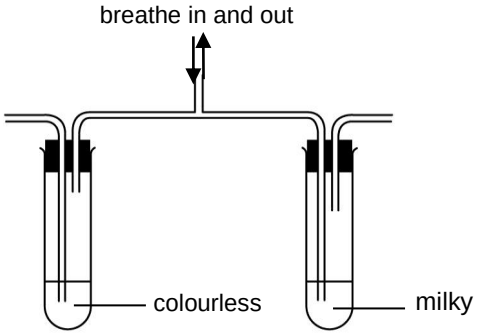
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Which of the following apparatus shows the results?

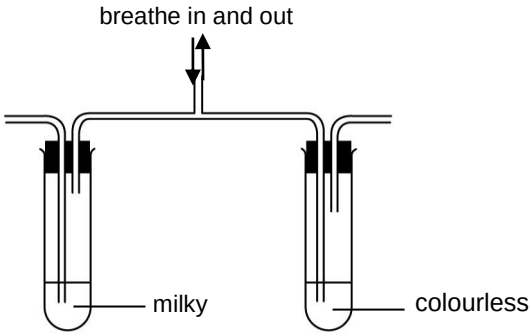
A.



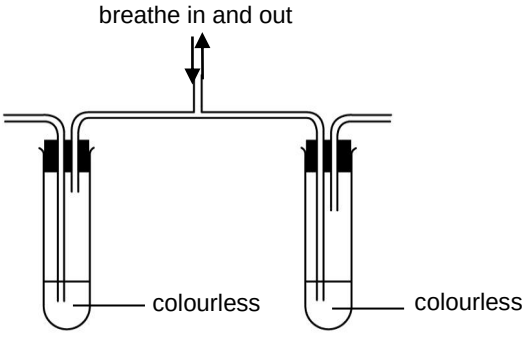
B.



C.



D.



□

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0080000043|!|

Which of the following is the correct pathway taken by air during inhalation?

A. air sacs → bronchi → bronchioles → trachea → larynx → nasal cavity
B. larynx → nasal cavity → bronchi → trachea → bronchioles → air sacs
C. nasal cavity → larynx → trachea → bronchi → bronchioles → air sacs
D. nasal cavity → trachea → larynx → bronchioles → bronchi → air sacs

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000044|!|

Which of the following is ***not*** a characteristic of the air sacs of human lungs?

A. The walls of the air sacs are made up of one-cell-thick epithelium.
B. The air sacs are surrounded by a network of capillaries.
C. The inner surface of the air sacs contains cilia to remove the dust particles.
D. The inner surface of the air sacs is moist.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000045|!|

Most of the carbon dioxide in the exhaled air

A. is produced inside the lungs.
B. originates from the atmosphere.
C. is a product of aerobic respiration.
D. is a product of anaerobic respiration.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0080000046|!| ★

What will happen if the inhaled air contains a lot of dust particles?

A. The exhaled air will contain a higher carbon dioxide content.
B. The rate of gas exchange will decrease.
C. The exhaled air will contain a lower oxygen content.
D. The rate of ventilation will decrease.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0080000048|!|

It is better to breathe through the nose than the mouth because

(1) the mouth does not lead to the trachea.
(2) the nose have hairs to filter out dust from the incoming air.
(3) the blood vessels in the nasal cavity can warm the incoming air.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0080000049|!| ★

Which of the following takes place to facilitate gas exchange during vigorous exercise?

A. The water film on the inner surface of the air sacs becomes thicker.
B. The walls of the air sacs become thinner.
C. The blood in the capillaries surrounding the air sacs flows faster.
D. More red blood cells are produced by the body.

##C##

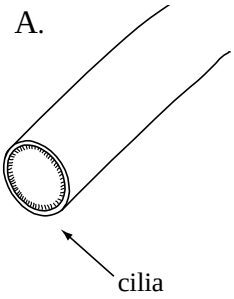
Skill assessed	Mark
Applying concepts	1

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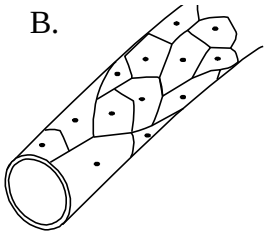
|!|BCEMC0080000050|!|

Which of the following tube-like structures transports inhaled air to the lungs? (The diagrams are not drawn to the same scale.)

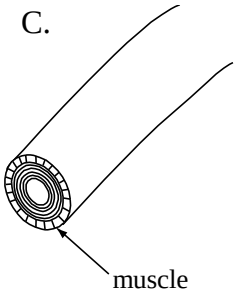
A.



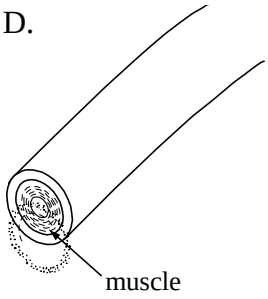
B.



C.



D.



☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0080000051|!| ★★

Under which of the following combinations would the diffusion rate of carbon dioxide from the blood capillaries to the air sacs be at the minimum?

	Amount of CO_2 in the blood in the capillaries	*Amount of CO_2 in the air in the air sacs*	*Rate of blood flow in the capillaries*
A.	small	large	low
B.	small	large	high
C.	large	small	low
D.	large	small	high

☐

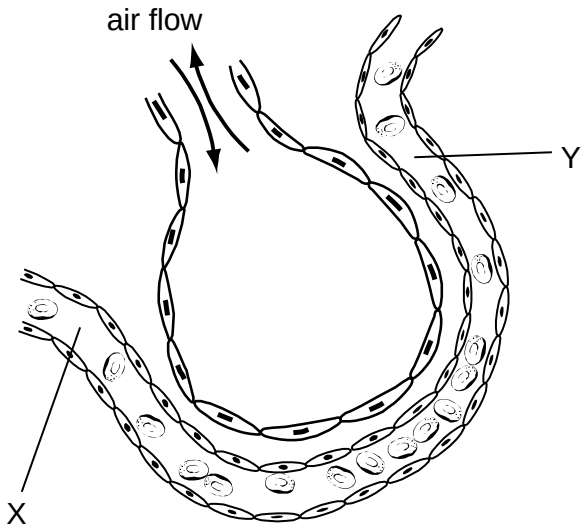
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0080000052|!| ★

The diagram below shows a tiny part of a human lung.



The carbon dioxide content at X is higher than that at Y.

Which of the following correctly shows the direction of blood flow and the blood vessel through which the blood flows back to the heart?

	Direction of blood flow	***Blood vessel***
A.	from X to Y	pulmonary vein
B.	from Y to X	pulmonary vein
C.	from X to Y	pulmonary artery
D.	from Y to X	pulmonary artery

□

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0080000056|!|

Which of the following series of changes occurs during inhalation?

	Intercostal muscles	*Diaphragm muscles*	*Volume of the thoracic cavity*
A.	relax	contract	decreases
B.	relax	relax	decreases
C.	contract	relax	increases
D.	contract	contract	increases

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000057|!| ★

At rest, our abdomen will bulge out when we inhale. This is because

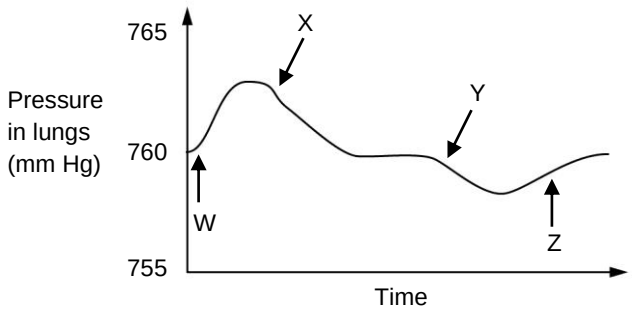
A. the downward movement of the ribs presses the abdomen outwards.
B. the flattening of the diaphragm pushes the abdominal organs outwards.
C. the abdominal muscles contract, pulling the abdominal organs outwards.
D. the pressure of the abdominal cavity increases during inhalation.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0080000058|!| ★★

The graph below shows the changes of the air pressure in the lungs during breathing. The atmospheric pressure is 760 mm Hg.



Which of the following positions on the graph corresponds to the point at which the ribs begin to raise?

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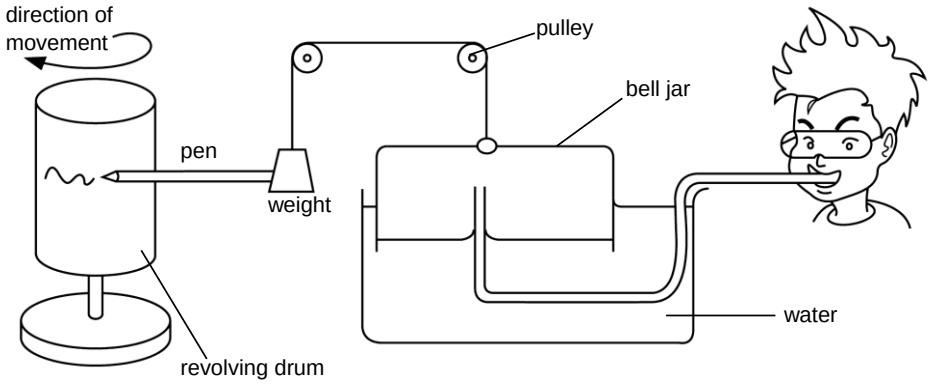
A. W
B. X
C. Y
D. Z

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000059|!| ★ SI

The following diagram shows a set-up used to investigate a person's breathing movement. It consists of an air-filled bell jar in which when the person breathes in and out, the bell jar moves up and down. This movement is recorded by the pen on the revolving drum.



What is the direction of movement of the pen when the person breathes in?

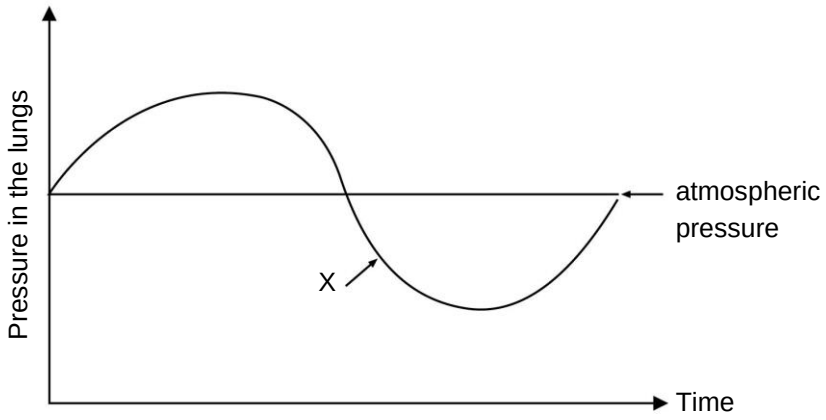
A. The pen moves up.
B. The pen moves down.
C. The pen does not move.
D. The pen moves anti-clockwise.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000060|!| ★

The graph below shows the pressure in the lungs of a man in one normal breath.



Which of the following combinations correctly shows what happens at time point X?

	Diaphragm muscles	*Ribs and sternum*	*State of ventilation*
A.	contract	move up	inhalation
B.	contract	move down	exhalation
C.	relax	move up	exhalation
D.	relax	move down	inhalation

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0080000061|!|

Ciliated epithelium can be found in

(1) the stomach.
(2) the trachea.
(3) the bronchi.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000062|!|

Which of the following statements about the lungs is ***incorrect***?

A. They are richly supplied with blood capillaries.

B. They are spongy.

C. They are surrounded by only one pleural membrane.

D. They are made up of many tiny air sacs.

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000063|!|

Which of the following statements about the air sacs in the lungs is ***incorrect***?

A. They have thin cell walls.

B. They are richly supplied with blood capillaries.

C. Their inner surface is kept moist by a water film.

D. They are the sites of gas exchange.

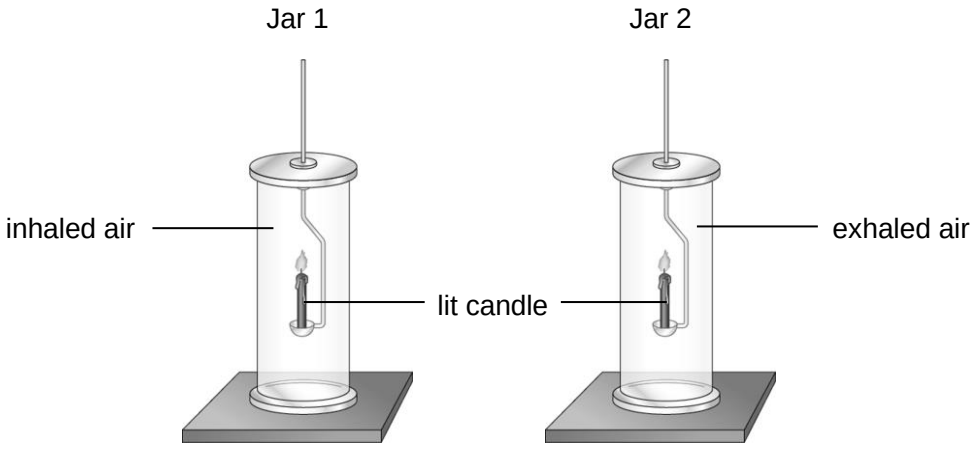
□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0080000065|!| ★★ **SI**

Directions: The following two questions refer to the diagram below, which shows an experimental set-up used to investigate the difference in composition between inhaled air (atmospheric air) and exhaled air.



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Which of the following would be observed through the course of the experiment?

A. The candle in jar 1 burns more vigorously than that in jar 2.
B. The candles in both jar 1 and jar 2 extinguish at the same time.
C. The candle in jar 2 extinguishes before the candle in jar 1.
D. More smoke is produced by the candle in jar 2.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0080000066|!| ★★ **SI**

Another jar, jar 3, is added to the experimental set-up. It contains exhaled air from an athlete who has just finished a 400-m race. Which of the following correctly states the expected observation and explanation?

	Expected observation	***Explanation***
A.	The candle in jar 2 will extinguish before that in jar 3.	The air in jar 2 contains more carbon dioxide than jar 3.
B.	The candle in jar 3 will extinguish before that in jar 2.	The air in jar 3 contains more carbon dioxide than jar 2.
C.	The candle in jar 2 will extinguish before that in jar 3.	The air in jar 2 contains less oxygen than jar 3.
D.	The candle in jar 3 will extinguish before that in jar 2.	The air in jar 3 contains less oxygen than jar 2.

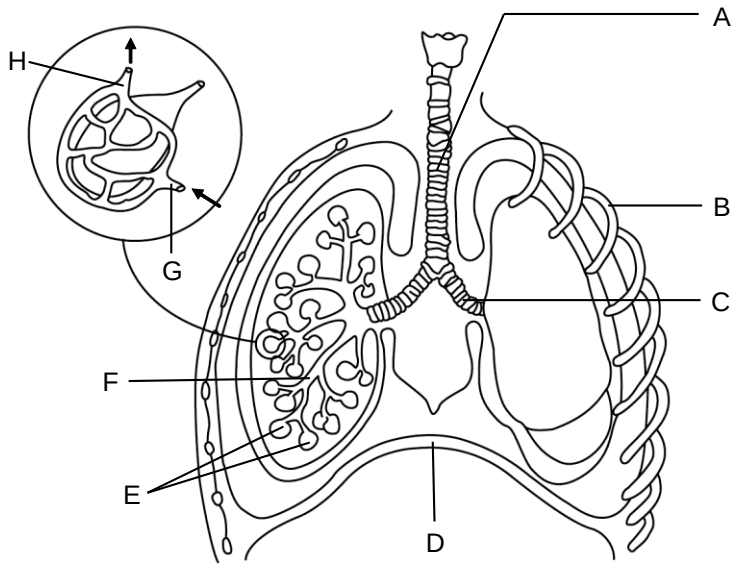
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0080000067|!|

Directions: The following three questions refer to the diagram below which shows part of the breathing system.



Which of the following structures is/are made up of C-shaped cartilage?

(1) A
(2) C
(3) F

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000068|!|

Which of the following are likely to affect the efficiency of gas exchange?

(1) the surface area to volume ratio of E
(2) the length of A
(3) the capillary density around E

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A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000069|!|

Which of the following statements about the breathing system is/are correct?

(1) The concentration of carbon dioxide at H is higher than that at G.
(2) B is made up of calcium phosphate.
(3) D becomes flattened during inhalation.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0080000070|!| ★★ SI

Directions: The following two questions refer to the following table which shows the oxygen and carbon dioxide concentrations in the atmospheric air, exhaled air collected at rest and exhaled air collected after exercise. Four samples are studied for each type of air.

Atmospheric air		***Exhaled air collected at rest***		***Exhaled air collected after exercise***	
O_2 (%)	CO_2 (%)	O_2 (%)	CO_2 (%)	O_2 (%)	CO_2 (%)
19.00	0.25	15.50	4.25	15.00	9.00
19.00	0.00	16.00	4.00	15.00	10.00
20.00	0.50	17.00	3.75	16.00	8.00
21.00	0.00	16.00	4.00	14.50	12.00

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Which of the following can be concluded from the results?

(1) The concentrations of oxygen and carbon dioxide are directly proportional.
(2) The carbon dioxide concentration in the exhaled air collected after exercise is higher than that in the exhaled air collected at rest.
(3) The oxygen concentration in the exhaled air collected after exercise is lower than that in the exhaled air collected at rest.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0080000071|!| ★★ **SI**

The concentration of carbon dioxide is found to be zero in some samples of the atmospheric air. Which of the following is/are the probable reason(s) for this?

(1) Carbon dioxide is absent in the atmospheric samples.
(2) The concentration of carbon dioxide is so low that it cannot be detected.
(3) The precision of the detecting instrument is low (data is rounded up to two decimal places only).

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

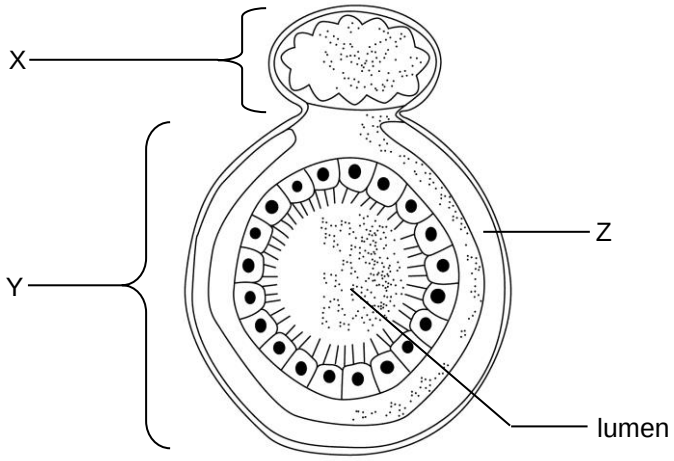
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMA0080000072|!| ★★ **Cross topic**

Directions: The following three questions refer to the diagram below which shows the transverse section of the trachea and the oesophagus.



Which of the following processes occur(s) in structure X?

(1) digestion
(2) absorption
(3) peristalsis

A. (1) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0080000073|!| ★★

Structure Y is finally branched to and terminated as

A. bronchi.
B. bronchioles.
C. air sacs.
D. blood capillaries.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0080000074|!| ★★

Why does the opening of structure Z face structure X?

A. To prevent the collapse of the trachea.

B. To increase the elasticity of the trachea.

C. To allow the oesophagus to stretch during swallowing.

D. To allow the trachea to stretch during breathing.

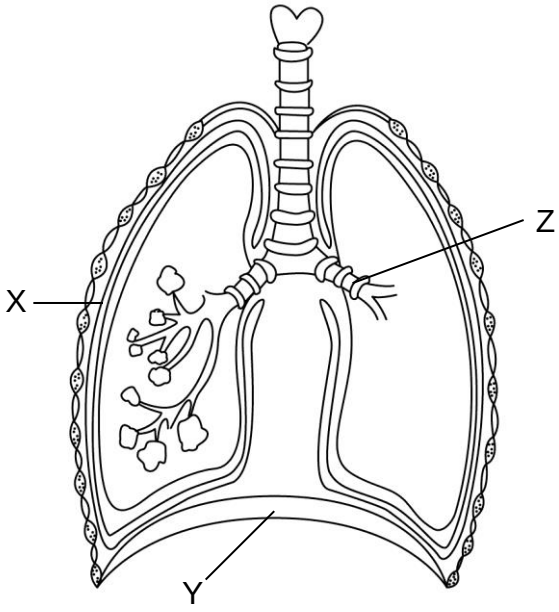
□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0080000075|!|

Directions: The following two questions refer to the diagram below, which shows part of the human breathing system.



Which of the following combinations correctly identifies structures Y and Z?

	Structure Y	***Structure Z***
A.	diaphragm muscle	bronchus
B.	intercostal muscle	bronchus
C.	diaphragm muscle	bronchiole
D.	intercostal muscle	bronchiole

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0080000076|!|

The function of the cells in structure X is

A. to secrete fluid which reduces friction between lungs and ribs during breathing.
B. to produce lymph to kill bacteria.
C. for gas exchange.
D. to secrete fluid which traps dust particles in inhaled air.

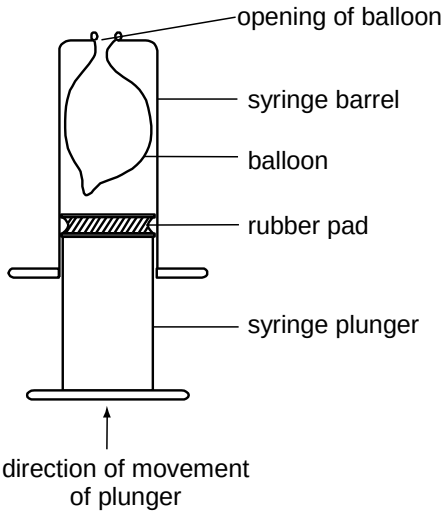
□

##A ##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0080000077|!| ★

Directions: The following two questions refer to the diagram below, which shows a model representing the thoracic cavity of a human. When the syringe plunger is pushed upwards, it represents a condition of ventilation.



The model demonstrates

A. how the changes in the pressure in the thoracic cavity affect the volume of the lungs.
B. how the diaphragm moves when we breathe.
C. how the rib cage moves when we breathe.
D. the pathway of air entering the lungs.

□

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0080000078|!| ★

What is the corresponding change in a real thoracic cavity when the syringe plunger is pushed upwards?

A. The rib cage moves outwards.
B. The diaphragm appears in a dome shape.
C. Air rushes into the lungs.
D. The intercostal muscles contract.

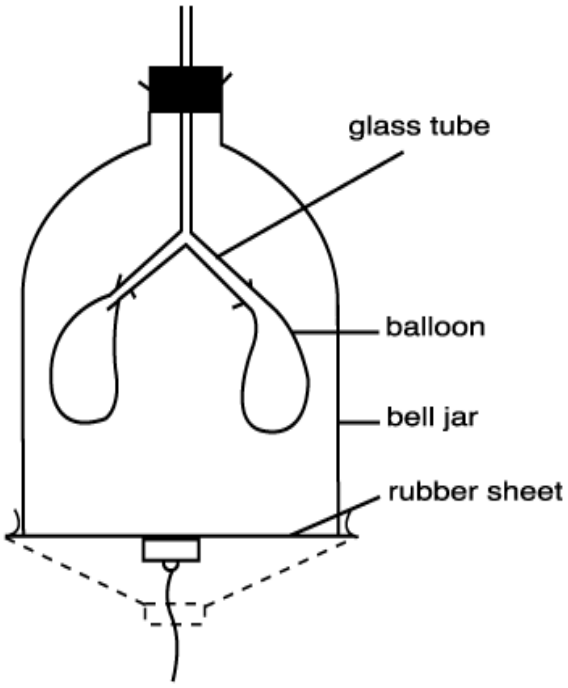
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000079|!| ★

Directions: The following three questions refer to the diagram below which shows a model demonstrating the human breathing movements.



Which of the following combinations correctly matches the structures of the breathing system with the various parts of the apparatus?

	Y-shaped glass tube	*Rubber sheet*
A.	trachea	thoracic wall
B.	trachea	diaphragm
C.	bronchus	thoracic wall
D.	bronchus	diaphragm

□

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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000080|!| ★

Which of the following correctly states and explains the results when the rubber sheet is pulled downwards?

	Volume of the balloons	***Reason***
A.	decreases	pressure inside the bell jar increases
B.	increases	pressure inside the bell jar increases
C.	increases	pressure inside the bell jar decreases
D.	no change	pressure inside the bell jar changes very little

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000081|!| ★ **SI**

This apparatus cannot truly represent the actual conditions in humans because

(1) the rubber sheet is pulled downwards by hand.
(2) the wall of the bell jar is rigid.
(3) there is only one opening in the model.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

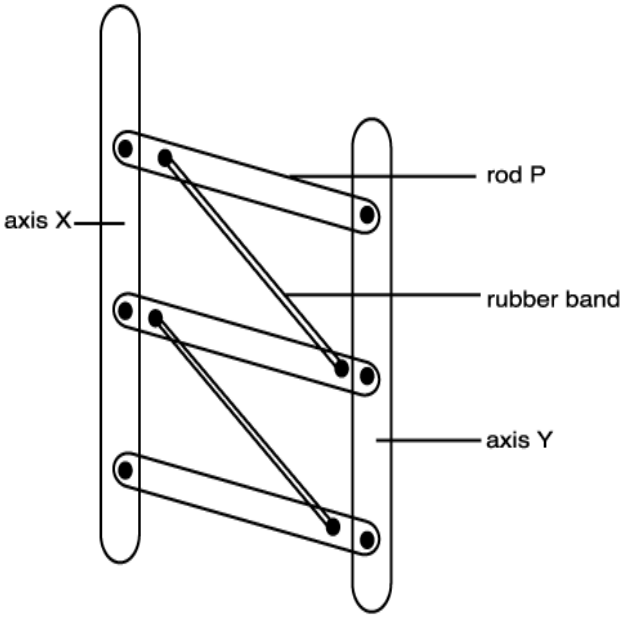
##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0080000082|!| ★

Directions: The following two questions refer to the diagram below which shows a rib cage model. Axis X is fixed and axis Y can be moved up and down.



Which of the following combinations correctly matches axis X, rod P and the rubber band with the structures in the human thorax?

	Axis X	***Rod P***	***Rubber band***
A.	vertebral column	intercostal muscle	rib
B.	vertebral column	rib	intercostal muscle
C.	sternum	intercostal muscle	rib
D.	sternum	rib	intercostal muscle

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0080000083|!| ★

Which of the following processes in the human body is demonstrated when axis Y is moved up?

A. respiration
B. inhalation
C. exhalation
D. gas exchange

□

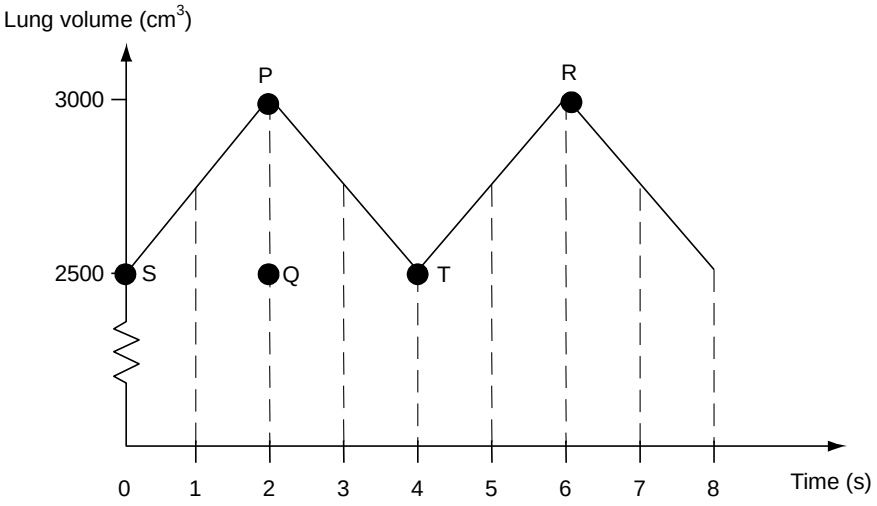
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0080000084|!| ★★

Directions: The following three questions refer to the graph below, which shows the changes in the lung volume of a person under normal conditions.



Inhalation occurs in

A. the 1st to 3rd second.
B. the 2nd to 4th second.
C. the 3rd to 5th second.
D. the 4th to 6th second.

☐

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0080000085|!| ★★

Which of the following combinations correctly shows the state of the intercostal muscles and diaphragm muscles at the 7th second?

	Intercostal muscles	*Diaphragm muscles*
A.	contract	relax
B.	relax	contract
C.	contract	contract
D.	relax	relax

☐

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0080000086|!| ★★

If the person is doing exercise, which of the following statements about the graph are correct?

(1) The distance between P and Q increases.
(2) The distance between P and R decreases.
(3) The distance between Q and T is greater than that between S and Q.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

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